



OpenMAX Sample Code Release Note

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Abstract

This document contains notes relating to the r0p0-00bet0 release of the OpenMAX sample code deliverables as supplied in the ARM deliverables bundle OX000-BU-000000-r0p0-00bet0.

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Feedback on this document

If you have any comments on or about this document, please send email to errata@arm.com giving:

- The document title
- The documents number
- The page number(s) to which your comments refer
- A concise explanation of your comments

General suggestion for additions and improvements are also welcome.

Change history

Issue	Date	Change
1.0	15 September 2006	Beta Release r0p0-00bet0

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1 PREFACE

This document contains general release information about the OpenMAX sample code r0p0-00bet0 deliverables and covers the following topics:

- Introduction
- Installation
- Build instructions
- Changes and fixes in this release
- Known issues

Intended audience

This document is written for all developers who are using the OpenMAX sample code software. It assumes that you are an experienced software developer, and that you are familiar with the development tools that you are using. It does not assume that you are familiar with the OpenMAX sample code software.

This document also assumes that the developer is familiar with the OpenMAX DL v1.0.1 API specification available from www.khronos.org

2 INTRODUCTION

2.1 OpenMAX DL

OpenMAX is a royalty-free, cross-platform API standard created and distributed by the Khronos Group, the same group responsible for the OpenGL and OpenGL-ES specifications.

The OpenMAX DL (Development Layer) standardizes access to a comprehensive set of low-level media processing primitives used extensively in audio, video and imaging applications.

The OpenMAX DL v1.0.1 standard is available free of charge from the Khronos website (www.khronos.org).

Currently in development, optimized implementation of the functions making up the OpenMAX API will be shipped with processors to enable library and codec implementers to rapidly and effectively make use of the full acceleration potential of new silicon - regardless of the underlying hardware architecture.

As well as general purpose media processing functions, OpenMAX DL also contains APIs specifically targeted for the implementation of codecs such as MPEG-4, H.264, MP3, AAC and JPEG.

2.2 OpenMAX DL Sample Code

This product is the ANSI C sample implementation of the functions described in the OpenMAX DL v1.0.1 API specification. All of the mandatory DL functions are supplied. None of the optional DLx API functions described in Appendix A of the specification are supplied.

This implementation attempts to be as mathematically accurate to the OpenMAX definition as possible, using floating-point code where necessary. It is not intended to reflect the expected performance or code size of an optimised (integer only) implementation.

This product is delivered as a source code only release. You will need to build the object libraries required to link to any application that makes use of the OpenMAX DL API. It is expected that you use the most suitable development tools for your hardware architecture in order to build these libraries.

OpenMAX DL is split into five application domains:

- AC - Audio Codecs (MP3 decoder and AAC decoder components)
- IC - Image Codecs (JPEG components)
- IP - Image Processing (Generic image processing functions)
- SP - Signal Processing (Generic audio processing functions)
- VC - Video Codecs (H264 and MP4 components)

2.3 Khronos OpenMAX header files

The following Khronos OpenMAX header files are provided as part of this release with the permission of the Khronos Group:

- ac/omxAC.h – Audio coding API
- ic/omxIC.h – Image Coding API
- ip/omxIP.h – Image Processing API
- sp/omxSP.h – Signal Processing API
- vc/omxVC.h – Video Coding API
- api/omxtypes.h – Definitions of OpenMAX DL basic data-types, structures, and error codes.

Consult the copyright notice in the above files for details on licensing and/or distribution restrictions.

2.4 Support

This product is supplied “as is”, and as such ARM offers no support for this product or its use. However, ARM does welcome any feedback on any functional issues with this product so that future releases may be improved. See the section on “Feedback on this Document” just before the table of contents.

Note: this is not intended to be an optimised implementation, it designed to be portable, simple to understand and accurate to the specification.

ARM will be releasing supported products containing optimised implementations of OpenMAX DL targeted at specific ARM architectures. Contact ARM for further details.

3 INSTALLATION

3.1 Files

The deliverables are collectively delivered as a single UNIX compressed tar file which has been gzipped. The download filename will be of the form:

`<download_name>.tgz`

Relocate the `<download_name>.tgz` file to an appropriate temporary location. Unpack the deliverable as follows, gunzip the *.tgz file and untar as follows:

```
gunzip <download_name>.tgz
```

```
gtar -xvf <download_name>.tar
```

Note GNU tar version 1.13 or later should be used to untar the deliverables as many versions of tar have problems dealing with very long path names. To find the version of gtar being used type `gtar --version`.

Note Under Windows, a GUI based application such as Winzip can be used to unpack the deliverables.

This will extract the deliverables into a directory named the same as the bundle OX000-BU-00000-r0p0-00bet0

The top level directory structures are shown in **Figure 2-1.** below

OX000-BU-00000-r0p0-00bet0

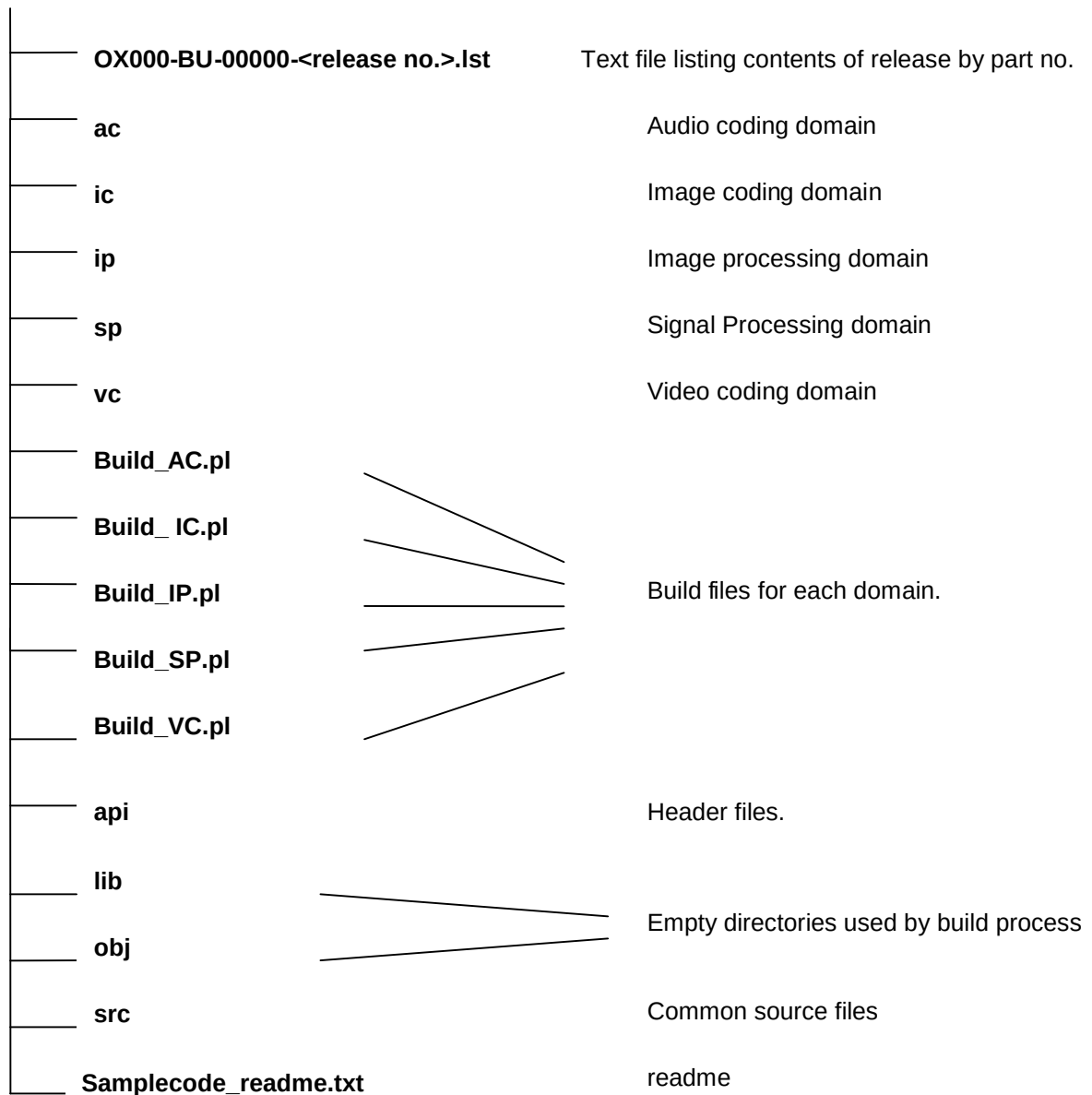


Fig 2.1 top-level directory structure

4 DOCUMENTATION

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at <http://www.adobe.com>. Select “Acrobat Reader” and download the reader for your computer platform/operating system.

4.1 OpenMAX sample code Errata

Not present in this release

4.2 Release Note

(This document)

This document contains general release information about the OpenMAX sample code product.

5 BUILD INSTRUCTIONS

The OpenMAX DL sample code has been written using ANSI standard C in order to be as portable across as many toolkits and target systems as possible.

This section goes through the steps required to build an object library for each of the five DL domains using the supplied example scripts which have been written to use the freely available GCC (GNU) toolkit.

5.1 Build instructions for GCC

A simple build script is provided for each of the domains. It requires:

- a Perl installation, version 5 or higher
- a GCC (GNU) tool chain

The build script is at the root of the source code tree. Invoke it using:

```
perl build_<domain-abbreviation>.pl
```

Where <domain-abbreviation> can be one of "AC", "IC", "IP", "SP", or "VC".

For example:

```
perl build_VC.pl
```

This will build a library archive called "omx<domain-abbreviation>.a". For example:

```
omxVC.a
```

It can be found in the "lib/" directory in the root of the source code tree.

5.2 Build instructions for other toolkits

If you have another tool chain then it is easy to modify the Perl build script to call that tool chain instead by editing the script to change the initialised values of the following Perl variables:

- \$CC – the Compiler command name
- \$CC_OPTS - the compiler options
- \$LIB – the Librarian command name
- \$LIB_OPTS - the librarian options
- \$LIB_TYPE - the file-type to be used for the output library

Once you have saved the Perl scripts then follow the instructions for the GCC build above.

6 DIFFERENCES FROM PREVIOUS RELEASES

This is the first release of OpenMAX sample code

7 KNOWN ISSUES

There are no known issues with this release